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ABSTRACT

To achieve high throughput and efficiency, semiconductor photolithography machines need an actuation system that can meet high acceleration and precision demands on the nanoscale. One available solution is the reluctance actuator, which provides higher acceleration and force output than the standard Lorentz actuator. A floating stage with air-bearings is used to eliminate friction in the photolithography process; however, vibration transfer is not entirely eliminated, leading to potential misalignment and asymmetries between the actuator elements. With asymmetrical offsets between mover elements, the output force can be greatly affected. This paper shows a method for estimating the force of various asymmetrical cases for the C-core reluctance actuator. Analytical models are developed and further improved through polynomial curve fitting using precomputed finite element simulation results from Comsol Multiphysics (COMSOL) to achieve more optimal solutions. An experiment verified the results of the force estimation equations, which were within ~11% for different cases of asymmetric air gaps. This contribution will lead to a design for a control system that will overcome the issue of asymmetries or other altered states.

I. INTRODUCTION

The production of integrated circuits (ICs) requires more advanced and optimized control for the future in order to increase IC throughput, memory storage, and processing speeds.[1,2] Photolithography machines are utilized to transfer and encode nanoscale geometric designs into the IC. To reduce vibration transfer, a floating stage with air-bearings is used, which eliminates friction.[1] These air-bearings allow the motion platform to hover, allowing no contact and, therefore, no friction or wear on nanoscale motions.[3] Vibration transfer is not entirely eliminated, however, since air vertices are generated within the air-bearing, inducing small vibrations into the positioning stage,[4] leading to self-excited instabilities that could damage the whole positioning stage.[5] This would lead to overall positioning inaccuracies, reducing the accuracy of the positioning stage.[6] Specifically with reluctance actuators, the misalignment would impart asymmetries, leading to significant differences in force output profiles depending on the asymmetry case. For this reason, it is important to investigate how the force can be estimated for various altered states.

High precision allows more geometric features to be encoded into ICs, leading to more encoded data.[7] In order to achieve high precision under larger strokes, a dual-stage actuation system is required.[2] This dual-stage system uses two actuation stages: a short-stroke stage and a long-stroke stage, which can achieve large range motions of up to 1 m with the long-stroke, and the motion can, then, be finely tuned on the nanoscale with the short-stroke stage; a schematic of the photolithography machine layout is shown in Fig. 1. Currently, Lorentz actuators are mostly utilized for the short-stroke in both the reticle and wafer stages. The reluctance actuator, however, can achieve higher force density and acceleration compared to the common Lorentz actuator.[7] Using reluctance actuators would lead to potentially better performing photolithography machines. That is why the reluctance actuator needs to be researched and modeled for all altered cases, leading to better motion control.

Lorentz actuators, which usually have a 1 mm stroke,[8] are easier to use since the input coil current and the output force are highly linearly dependent, leading to concise models of the force estimation.[9] This allows the photolithography machines to operate with a high

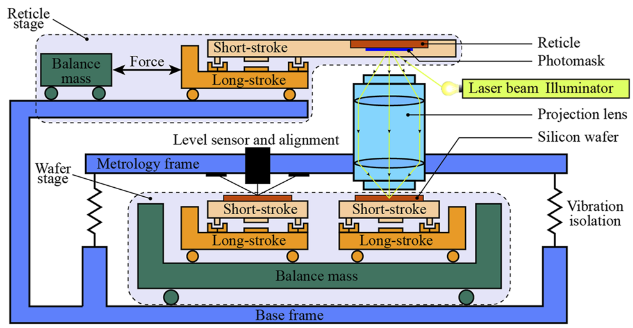

FIG. 1. Basic principle of the lithography process: A schematic representation of the lithography machine architecture, which consists of three main parts: (i) the reticle stage, (ii) the projection lenses, and (iii) the wafer stage. Both the reticle stage and the wafer stage use a dual-stage long-stroke and short-stroke using Lorentz actuators to correct positioning errors.

positioning accuracy over large ranges.[10] The downside to Lorentz actuators is the force density that is relatively low, meaning that significant current is required, which leads to much power being wasted in the form of heat requiring the use of complex water-cooling systems.[9] The reluctance actuator, however, is much more efficient in the consumption of current, and there is evidence that it can yield 10 times the force density of the Lorentz actuator.[9]

The reluctance actuator is not as easy to model as the Lorentz actuator since the current–force and the position–force relationship is nonlinear and needs to be evaluated.[9] There have been many contributions to examining these nonlinearities in the literature. In Ref. 11, for instance, the nonlinear relationships are modeled for real-time control applications with the use of feedback from a capacitive displacement sensor, attaining 2 nm resolution and a range of 60 μm.

There have also been many instances of photolithographic control designs published. For example, in Ref. 12, a synchronization control algorithm is designed to improve synchronization between the reticle and wafer stages. In Ref. 13, a combined system identification with robust control is showed for high-precision motion control systems. In Ref. 14, a data-based multi-input multi-output (MIMO) feedforward control method is applied to the motion control of the wafer stage, which has the ability to counteract unknown disturbances in the motion stage. In Ref. 15, feedforward control is implemented using known data for the wafer stage to improve the settling time and minimize crosstalk in the system.

Several studies have considered the reluctance actuator modeling and control strategies in the literature. For example, the accuracy of finite element analysis (FEA) is examined in Ref. 16 by simulating the frequency response of a two-DoF hybrid reluctance actuator under monoaxial excitation, which was, then, verified through experiments. A flux-controlled hybrid reluctance actuator is explored in Ref. 17, where magnetic flux is an input, and sensor fusion of a search coil and a current monitor is used to predict flux with the nonlinearities being controlled with the use of a PI flux feedback controller. A flux feedback-controlled reluctance actuator is also examined in Ref. 18, with a cascaded analog sensing coil and a hall probe flux feedback, and the reluctance actuator could be linearized and verified through experiments. This gives the reluctance actuator the linear behavior of the Lorentz actuator along with the advantage of greater energy efficiency. Flux leakage can also be a problem, and having a good detection method is important. In Ref. 19, a compensation algorithm for magnetic flux leakage measurement errors is defined and verified through simulation and physical experiments. In Ref. 20, a run-to-run control based on Bayesian optimization of a reluctance actuator's input current is presented, which can be used for soft landing control and verified experimentally. In addition, modeling the hysteresis effects of the reluctance actuator can also enhance the model. In Ref. 21, a nonlinear current control configuration is proposed with hysteresis compensation by using an adaptive multilayer neural network. A regressive radial basis function (RBF) neural network model for rate-dependent hysteresis modeling achieves high performance in hysteresis estimations for reluctance actuators when compared to other methods, as outlined in Ref. 22. The influence of hysteresis is also modeled in Ref. 23 with the use of a hysteretic inverse actuator model[23] and used to linearize a current-driven reluctance actuator. In Ref. 24, two semi-analytic methods are evaluated to predict magnetic hysteresis in the output force of an E-core reluctance actuator. In Ref. 25, an open-loop steady-state model is examined, which can predict the hysteresis effects of an E-core actuator based on the phase delay of flux variations. A hybrid lumped-parameter model is given in Refs. 26 and 27, which considers the mechanical and electromagnetic dynamics, along with eddy currents, flux fringing, magnetic hysteresis, and saturation. Computed Comsol Multiphysics (COMSOL) data are used in Ref. 27 to model the induced eddy currents and flux fringing effects. Fault diagnosis is another area that needs more research. There has been fault diagnosis conducted for a switched reluctance motor in Ref. 28 but nothing yet for the reluctance actuator.

This paper examines asymmetrical cases of the reluctance actuator from angular disturbances, as outlined in Fig. 2. There is an example of case 1 being examined in Ref. 7 with the use of a block diagram for disturbance simulation being briefly outlined. In Ref. 29, an experimental testbed is presented, which can impart case 1 disturbances with the use of piezoelectric actuators to test out algorithms for disturbance control. There has not been an in-depth analysis on a reluctance actuator with disturbance cases in all rotational degrees of freedom, as in Fig. 2.

This paper outlines how angular disturbance cases in each degree of freedom of a C-core reluctance actuator affect the output force. The analytical solutions are further optimized by comparing them to COMSOL FEA models to generate polynomial correction factors for each case. In addition, a filleted C-core is modeled using polynomial correction factors. Each case is compared to

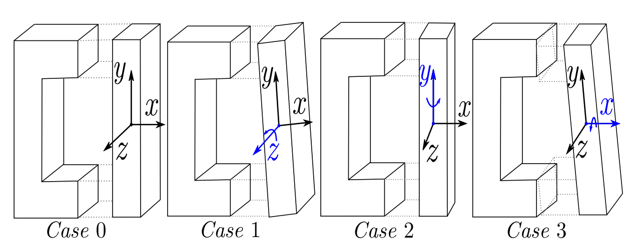

FIG. 2. Case 0 is the ideal case with symmetrical air gaps, whereas cases 1, 2, and 3 have angular disturbances in the *z*, *y*, and *x* axis, respectively.

experimental findings. This paper is outlined as follows: an analytical model derivation for each disturbance case is presented in Sec. II. The COMSOL FEA models are built and compared to the analytical model in Sec. III, and a polynomial correction function is found. In Sec. IV, the experimental apparatus is outlined, and the results are shown along with comparisons to the analytical results with a discussion presented in Sec. V. Finally, Sec. VI shows the conclusions of this study and the future work.

II. ANALYTICAL MODEL

The ideal symmetrical case is defined as case 0, whereas cases 1, 2, and 3 have angular disturbances about the z, y, and x axis, respectively. There are forces along the I-beam pushing it toward the C-core element from Fig. 3. For case 1, asymmetries result in different forces in the top and bottom, inducing torque onto the I-beam. There would also be flux fringing and hysteresis effects, as shown in Ref. 26, but it is not evaluated in this paper. The reluctance for a non-parallel gap is found through integration of the gap along the edge. The analytical model provides a rough estimate of whether the force will be and will not be accurate in some cases, so FEA modeling is used to further the force estimation through the use of polynomial correction factor functions to match the FEA data found.

A. Case 0 derivation

This case is for the standard symmetrical air gap scenario and can be found in the literature; see, for example, Refs. 7, 9, 11, 17, 18, and 24–26. However, most of these references do not take into full account the mean path length, l_m, of the magnetic material. With the assumption that no fringing occurs, the flux density, B, can be found along with the mean path length, l_m,

$$l_m = 2(h + w - b), \quad B = \frac{\mu_0 N i}{\frac{l_m}{\mu_r} + 2a}, \tag{1}$$

where μ_0 is the permeability of free space, μ_r is the relative permeability of the C-core and I-beam material, N is the number of coil turns, i is the coil current, l_m is the mean path length, and a is the air gap displacement.

Gauss's law states that the flux entering and leaving a surface is equivalent,[30]

$$\oiint B * dS = 0. \tag{2}$$

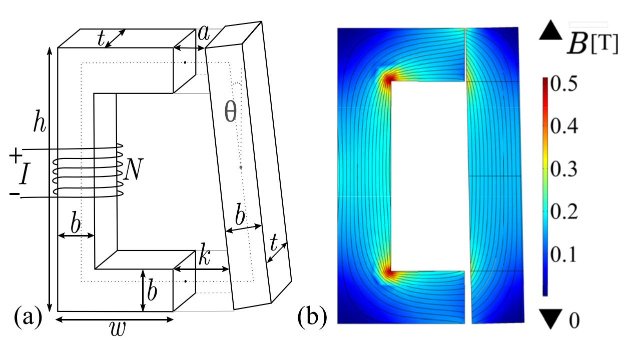

FIG. 3. Case 1: (a) Schematic representation of the C-core reluctance actuator and (b) COMSOL flux density distribution FEA of the C-core.

The Maxwell stress tensor can reduce to[7]

$$F = \frac{1}{2\mu_0}\int_S B^2 dS = \frac{A}{2\mu_0}B^2 = \frac{\mu_0 A N^2 i^2}{2\left(\frac{l_m}{\mu_r} + 2a\right)^2}, \tag{3}$$

where F is the force in each air gap (F_1 is the force in the upper air gap and F_2 is the force in the lower air gap; for this case, $F = F_1 = F_2$) and A is the cross-sectional area.

B. Case 1 derivation

In case 1, the I-beam is subjected to a rotation in the z axis resulting in asymmetric tilted air gaps, as shown in Fig. 4.

$$g(y) = a + (b - y)\tan(\theta). \tag{4}$$

Then,

$$dP_1(y) = \frac{-\mu_0 t dy}{\frac{c}{b}y - (a + c)}, \tag{5}$$

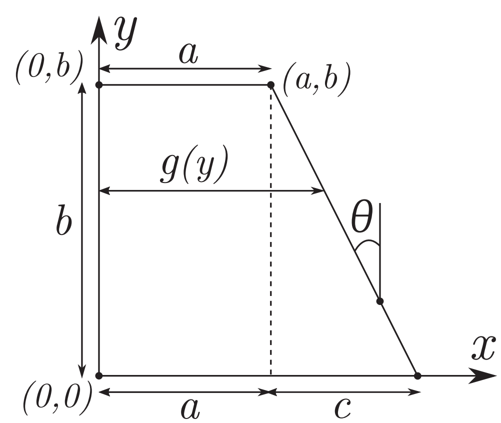

FIG. 4. Geometric representation of the tilted air gap for the case 1 reluctance derivation with a rotation in the z axis of the I-beam.

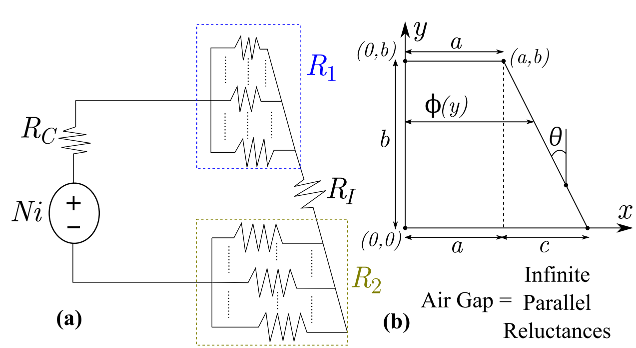

FIG. 5. The reluctance actuator is modeled as a magnetic circuit (a) with the air gaps being represented (b) as a set of infinite reluctances in parallel.

and we can write

$$P_1 = \int_0^b \frac{-\mu_0 t}{\frac{c}{b}y - (a+c)}\,dy = \frac{\mu_0 A}{b\tan\theta}\ln\left(\frac{b\tan\theta}{a} + 1\right) \tag{6}$$

and

$$R_1 = \frac{1}{P_1} = \frac{b\tan\theta}{\mu_0 A\,\ln\left(\frac{b\tan\theta}{a} + 1\right)}, \tag{7}$$

where P_1 is the permeance of the top air gap, which is the reciprocal of the reluctance R_1. The reluctance in the second gap, R_2, can be found by substituting a for $k = a + (h - b)\tan\theta$ as

$$R_2 = \frac{b\tan\theta}{\mu_0 A\,\ln\left(\frac{b\tan\theta}{k} + 1\right)}. \tag{8}$$

The reluctance in the C-core, R_C, and the I-beam, R_I, is found from the geometry as

$$R_C = \frac{h + 2(w - b)}{\mu_r\mu_0 A}, \quad R_I = \frac{h}{\mu_r\mu_0 A\cos\theta}. \tag{9}$$

Then, the total reluctance of the system is then found to be the summation of all the reluctances since they are in series as follows:

$$R_T = R_1 + R_2 + R_C + R_I. \tag{10}$$

Then, based on (3), the force, F, through a section of area, A, in terms of the magnetic flux, $\phi = BA$, can be defined as

$$F = \frac{\phi^2}{2\,\mu_0 A}. \tag{11}$$

The flux through the air gaps is modeled as a set of infinite parallel paths as illustrated in Fig. 5. Therefore, the flux can be defined as

$$d\phi_1(y) = \frac{Ni}{R_T}R_1 dP_1(y) = \frac{-NiR_1\mu_0 t dy}{R_T\left(\frac{c}{b}y - (a+c)\right)}. \tag{12}$$

Combining (11) and (12) yields

$$dF_1(y) = \frac{d\phi_1^2}{2\,\mu_0 t dy} = \frac{N^2 i^2 R_1^2 \mu_0 t dy}{2R_T^2\left(\frac{c}{b}y - (a+c)\right)^2} \tag{13}$$

and

$$F_1 = \int_0^b \frac{N^2 i^2 R_1^2 \mu_0 t}{2R_T^2\left(\frac{c}{b}y - (a+c)\right)^2}\,dy = \frac{N^2 i^2 R_1^2 \mu_0 A}{2R_T^2 a(b\tan\theta + a)}, \tag{14}$$

where F_1 is the force in the top air gap. The force in the bottom air gap, F_2, is found similarly by replacing a with k and R_1 with R_2,

$$F_2 = \frac{N^2 i^2 R_2^2 \mu_0 A}{2R_T^2 k(b\tan\theta + k)}. \tag{15}$$

C. Case 2 derivation

In case 2, the rotation of the I-beam about the y axis provides equal tilted air gaps in the upper and lower gap sections, as shown in Fig. 6. The geometric representation for case 2 follows the same layout as case 1, assuming $t = b$. The upper and lower gaps are the same for case 2; therefore, the reluctance in both air gaps is the same, resulting in the same force produced. Using a similar derivation from case 1, the upper and lower reluctance can be found,

$$R_1 = R_2 = \frac{b\tan\theta}{\mu_0 A\,\ln\left(\frac{b\tan\theta}{a} + 1\right)}, \tag{16}$$

leading to the forces as follows:

$$F_1 = F_2 = \frac{N^2 i^2 R_1^2 \mu_0 A}{2R_T^2 a(b\tan\theta + a)}. \tag{17}$$

D. Case 3 derivation

Case 3 is similar to case 0, where the air gaps are symmetrical. However, the projected area that the magnetic flux flows through is changed with the rotation in the x axis, as shown in Fig. 7.

For case 3, the projected gap area, A_G, needs to be determined. This can be done, as shown in Fig. 7, through the subtraction of A_s and A_T from the original as

$$A_G = b^2 - A_s - A_T. \tag{18}$$

The geometric parameters can then be found through trigonometry as

$$\chi = \frac{1}{2}(h - b\sin\theta)\tan\theta, \tag{19}$$

$$\varepsilon = \frac{1}{2}(h - b(2 + \sin\theta))\tan\theta, \tag{20}$$

$$\lambda = \frac{h}{2}(1 - \sec\theta) - \frac{b}{2}\tan\theta, \tag{21}$$

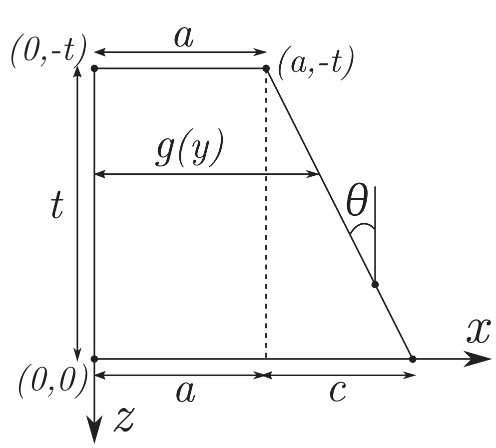

FIG. 6. Geometric representation of the tilted air gap for the case 2 reluctance derivation with a rotation in the y axis of the I-beam.

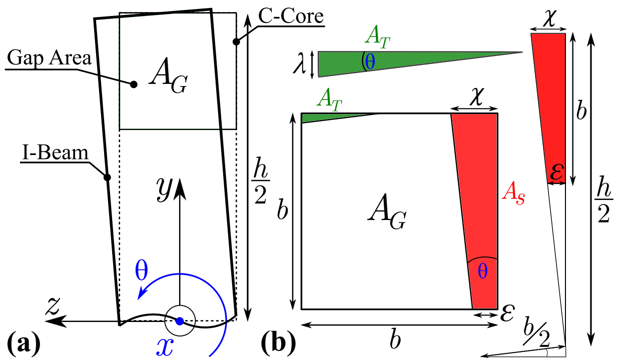

FIG. 7. Geometric representation (a) of the tilted cross section for the case 3 reluctance derivation with a rotation in the x axis of the I-beam. The air gap cross section (b) is shown to not be square due to the tilted I-beam.

$$A_T = \frac{\lambda^2}{2\tan\theta}, \quad A_s = \frac{\chi^2 - \varepsilon^2}{2\tan\theta}. \tag{22}$$

The gap area can then be found through (16) as

$$A_G = b^2 - \frac{\chi^2 - \varepsilon^2 - \lambda^2}{2\tan\theta}. \tag{23}$$

Since the rotation is in the middle of the I-beam, the gap area will be equal due to symmetries. This leads to the force equation, such as in case 0; the area, A, is replaced by the gap area, A_G, as

$$F_1 = F_2 = \frac{N^2 i^2 \mu_0 A_G}{8\left(\frac{h+w-b}{\mu_r} + a\right)^2}. \tag{24}$$

This equation estimates the force more accurately for smaller air gaps. With a bigger air gap, the flux will have room to travel, and the gap area, A_G, would be more similar to the standard area, A, from case 0. However, the polynomial correction factor found in Sec. III fixes this issue for all cases and allows the force to match for all air gaps and angles.

III. SIMULATION RESULTS

In order to verify the analytical model, the reluctance actuator was built and verified using COMSOL, which is a finite element analysis (FEA) Multiphysics modeling software engine. A similar approach was conducted in Refs. 26 and 27 by using FEA to find physics relationships for complex systems to obtain approximate governing equations. The reluctance actuator to be examined feature dimensions of $b = t = 25$ mm, $w = 60$ mm, $h = 140$ mm, and $N = 400$ turns for all simulation and experimental models as based on the previous schematic in Fig. 3(a).

The model material consists of Hiperco 50, annealed at 1411 °F, which is an iron–cobalt vanadium alloy. The magnetic field density vs the magnetic field intensity curve or commonly known as the BH-curve was utilized, as shown in Fig. 8. The analytical model can then be corrected, considering this saturation using a lookup table of the given BH-curve of the material used when solving the derived equations.

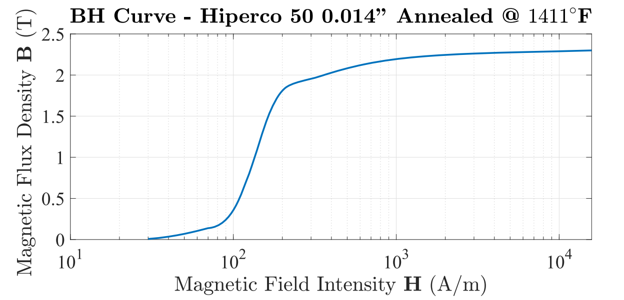

FIG. 8. BH curve for Hiperco 50 0.014 in. annealed at 1411 °F in which the I-beam and C-core are comprised of. This curve is used in the simulation and analytical models.

A. Case 0

First, case 0 is built and tested, as shown in Fig. 9. Using the analytical solution from (3), the forces are found and compared to the COMSOL model with the current set to $i = 1$ A and the number of turns set to $N = 400$ for a gap range of $a = 0.25$–2 mm. The results match very closely with a maximum error of 1.26% between models, as shown in Fig. 10.

These results show good matching between COMSOL and the analytically derived solution. Next, the other cases will be evaluated with COMSOL to determine the simulated force responses.

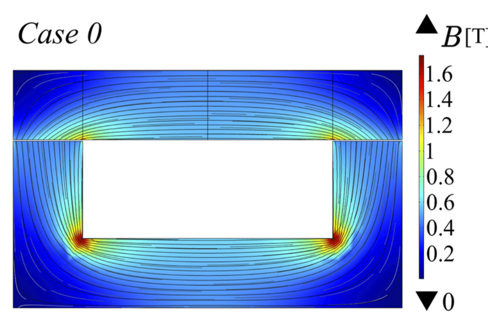

FIG. 9. COMSOL model of the case 0 reluctance actuator for an air gap separation of 0.5 mm.

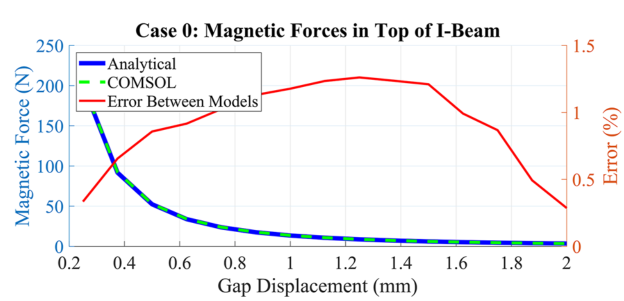

FIG. 10. The analytical and COMSOL models for case 0 are shown to match with a maximum error of 1.26% between forces in the top air gap with a direct current of 1 A and 0.25–2 mm of gap separation.

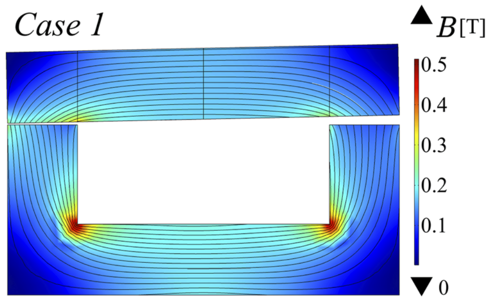

FIG. 11. COMSOL model for the case 1 reluctance actuator for an air gap separation of 0.5 mm and an angle of 1.2° in the z axis.

B. Case 1

The next case is modeled in COMSOL and compared with the analytical model based on (14) and (15). When running the simulation, the area that the flux passes through is different from the original assumption since there is an angle and the flux is more concentrated near the area where the I-beam and C-core are close as shown in Fig. 11 and does not flow in a linear fashion. Due to the physical complexity of the asymmetry, the model needs to be corrected using a 3D surface map correction factor based on the angle and gap separation to provide better matching. The 3D correction map does not consider any approximations or assumptions, and a polynomial equation for the effective area can be derived from the correction map with curve fitting tools to better match the ideal flux path.

The 3D surface of the COMSOL force vs z-angle and gap separation was generated and was compared to the analytical solution through a ratio and fitted to a 3D polynomial curve using MATLAB, as shown in Fig. 12. The correction factors to the top and bottom areas were, then, determined in order to 3D curve fit the data to the data from COMSOL. The new force equation which is simply Eq. (14) and Eq. (15) with the original area, A, replaced by $\boldsymbol{A}_1^*$ and $\boldsymbol{A}_2^*$, found in Eq. 25, for the top [Eq. (14)] and bottom [Eq. (15)] forces, respectively. This solution also works well with various current, as shown in Fig. 13.

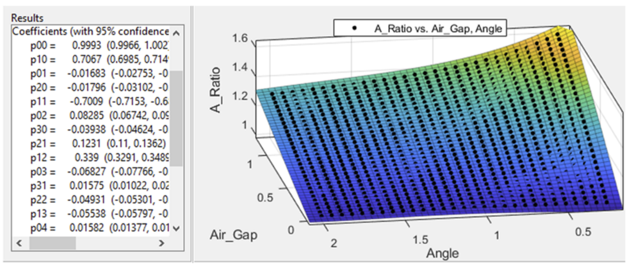

FIG. 12. Comparing the COMSOL results with the analytically derived equation for the top force and fitting the ratio to a 3D surface with the use of MATLAB's curve fitting toolbox, the correction factor equation for the new area can then be derived. The same is produced for the bottom force equation.

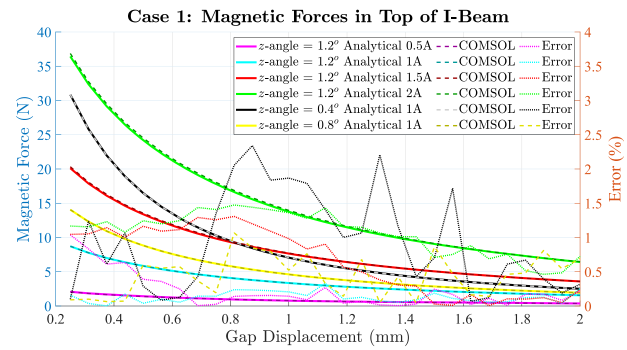

FIG. 13. The COMSOL and analytical model of the top gap force of case 1 are compared for various current ranging from 0.5 to 2 A and z-angle disturbances. This confirms that the equations work for various current inputs with the biggest error being 2.3%.

$$
\begin{aligned}
\boldsymbol{A}_1^* = \big(&0.9993 + 0.7067\theta - 0.016\,83a - 0.017\,96\theta^2 - 0.7009\theta a \\
&+ 0.082\,85a^2 - 0.039\,38\theta^3 + 0.1231\theta^2 a + 0.339\theta a^2 \\
&- 0.068\,27a^3 + 0.015\,75\theta^3 a - 0.049\,31\theta^2 a^2 - 0.055\,38\theta a^3 \\
&+ 0.015\,82a^4\big)A, \\
\boldsymbol{A}_2^* = \big(&0.9891 - 0.1223\theta + 0.025\,04a + 0.1897\theta^2 - 0.2571\theta a \\
&+ 0.022\,04a^2 - 0.092\,27\theta^3 + 0.028\,59\theta^2 a + 0.1626\theta a^2 \\
&- 0.032\,98a^3 + 0.033\,01\theta^3 a - 0.035\,44\theta^2 a^2 - 0.024\,64\theta a^3 \\
&+ 0.008\,659a^4\big)A.
\end{aligned} \tag{25}
$$

C. Case 2

For case 2, the COMSOL model was built in 3D as shown in Fig. 14 and, then, tested and compared to analytical results through various y-angles to determine the polynomial correction factors following the same procedures as in case 1. For this case, the forces in the top and bottom were found to be approximately the same since it behaves symmetrically. The correction factor would replace the area, A, in Eq. 17 by $\boldsymbol{A}^*$,

$$
\begin{aligned}
\boldsymbol{A}^* = \big(&0.9837 + 0.006\,107\theta + 0.072\,28a - 0.006\,254\theta^2 \\
&+ 0.008\,648\theta a - 0.034\,65a^2 + 0.000\,508\,2\theta^2 a \\
&- 0.003\,761\theta a^2 + 0.005\,482a^3\big)A.
\end{aligned} \tag{26}
$$

D. Case 3

For case 3, the COMSOL model was built in 3D and tested as shown in Fig. 15 and then compared to analytical results through various x-angles in order to find the polynomial correction factors on the area. Similar to case 2, the forces in the top and bottom air gaps were found to be the same as also found in the original analytical derivation. The polynomial correction can, then, be applied by replacing area, A_G, with $\boldsymbol{A}_G^*$,

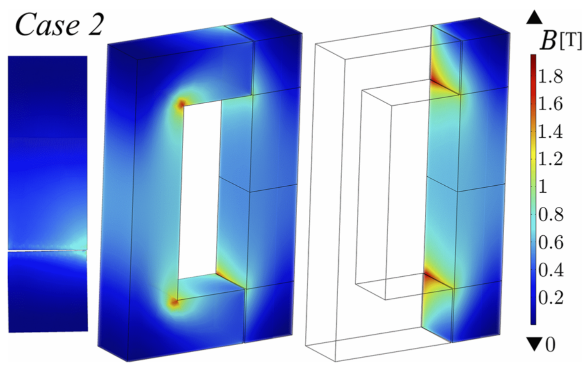

FIG. 14. COMSOL model for case 2 with a gap separation of 0.25 mm and an angle of 1.2° in the *y* axis.

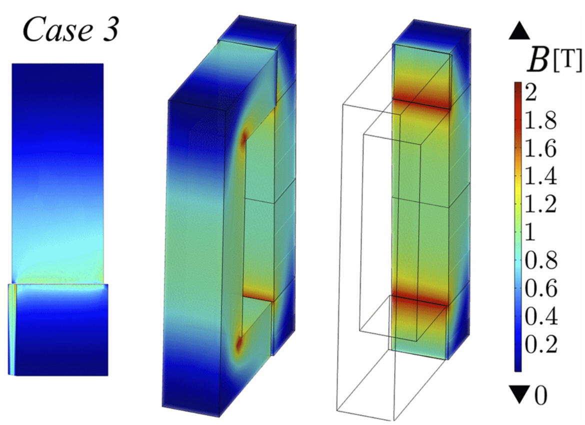

FIG. 15. COMSOL model for case 3 with a gap separation of 0.25 mm and an angle of 1.2° in the *x* axis.

$$A_G^* = \left(0.9812 + 0.023\,93\theta + 0.071\,14a - 0.016\,78\theta^2 + 0.012\,81\theta a - 0.022\,72a^2\right) * A_G. \quad (27)$$

E. Case 0 F

For case 0 F, fillets are introduced to observe whether fillets can help improve the actuator performance. The case is the same as case 0; however, 5 mm fillets are introduced at the edges of the C-core, as shown in Fig. 16.

The COMSOL results are, then, compared to case 0 results, and a polynomial correction factor, A_f^*, to replace the area, A, is found,

$$A_f^* = \left(0.007\,16a^3 + -0.0586a^2 + 0.175a + 0.675\right)A. \quad (28)$$

The analytical cases can, then, be compared to case 0 to determine their effect, as shown in Fig. 17.

IV. EXPERIMENTAL DESIGN

In order to analyze the forces in the I-beam at various air gaps, two enclosures and a force bracket were built in SolidWorks and

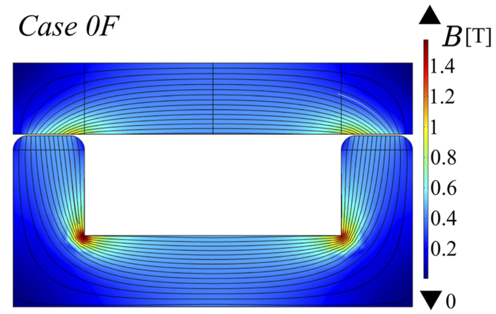

FIG. 16. COMSOL model of the case 0 F reluctance actuator, which features 5 mm fillets, for an air gap separation of 0.5 mm.

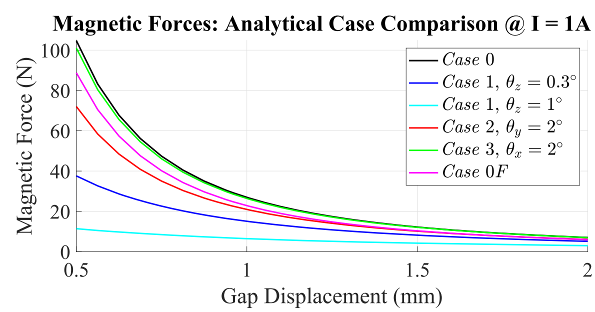

FIG. 17. Total analytical I-beam output force comparison of all considered cases at a current of I = 1 A and a 0.5–2 mm of gap separation.

3D printed. These enclosures will isolate and house the I-beam and the C-core separately, allowing for static measurements of the force induced at various gaps. These enclosures are shown in Fig. 18. The I-beam and C-core were built using layered laminations of the annealed Hiperco 50 iron–cobalt vanadium alloy.

The I-beam is seated inside of the rectangular slot of the I-box and fixed with two aluminum pins that goes through both the I-beam and the I-box. The pin goes through a region with minuscule

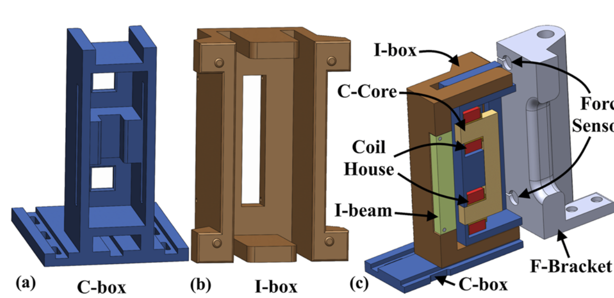

FIG. 18. The (a) C-box and (b) I-box house the C-core and I-beam, respectively. The F-bracket (c) houses the force sensors and is what the I-box pushes against. Cross-sectional model (c) of SolidWorks of the experimental assembly is also shown. All housing components are 3D printed.

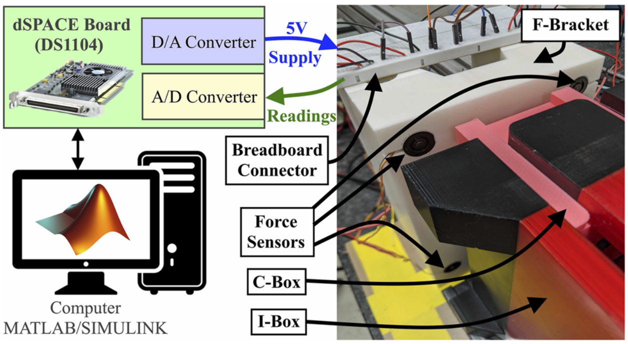

FIG. 19. The experimental setup is shown with the 3D printed components F-bracket, C-box, and I-box along with the four force sensors connected to the breadboard, which is, then, connected to the dSPACE board (DS1104), which supplies the 5 V input to the sensors and reads the sensor data through A/D converters, which are sent to the computer through MATLAB Simulink.

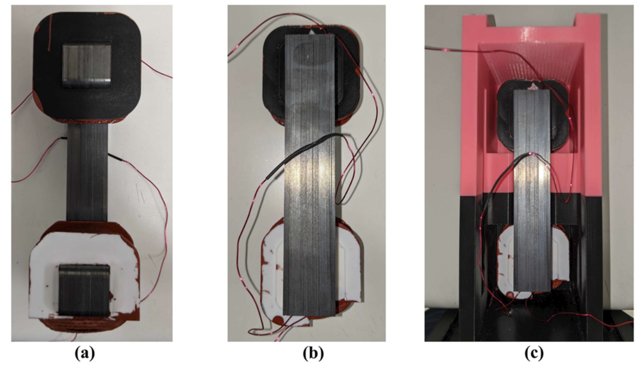

FIG. 20. The (a) front and (b) back of the C-core element with the coils and an inside view (c) of the C-core inside the C-box.

flux contribution, therefore not greatly affecting any of the calculations. From the analysis, at a 0.25 mm gap, the force in the top and bottom gap without the holes was found to be 201.57 and 201.49 N, respectively, whereas with the holes, the forces were 201.54 and 201.44 N, respectively. Therefore, it was found that the holes do not affect the force in any substantial way.

There are two coils in this design that are seated on the two teeth of the C-core. Each coil has a 3D printed coil house in which the coil wire wraps around. Each coil features 200 turns, giving $N = 400$. These coil houses are, then, slotted onto the C-core, which is, then, seated into the C-box. The force bracket or F-bracket is designed to be mounted behind the C-box. The force from the I-box is pressed up against and measured through four compression sensors (FX293X-100A-0010-L)[31] that are connected to the vertical divots, as shown in Fig. 18. The I-beam is attracted to the C-core, therefore pushing the I-box, which, then, pushes against the F-bracket. The dimple extrusions of the I-box will press against the sensors. Each of the four sensors was calibrated individually with various weights and plotted to find out the sensitivity and offset required based on the voltage readings. Each sensor had slightly different sensitivities being 114.05, 115.38, 114.48, and 114.91 mV/N, with an offset of −5.53, −5.25, −5.09, and −5.51 mV, respectively. The experimental apparatus is shown in Figs. 19 and 20. These sensors are connected to a dSPACE DS1104 control board, which reads the data and based off the voltage reading and the previous sensitivity analysis, and the output force can, then, be evaluated.

To position the I-beam in various angular disturbance cases, a modular I-box design was built where the backing can be switched out for different cases and 3D-printed, as shown in Fig. 21.

For case 0 F, a filleted C-core is utilized as shown in Fig. 22(b) along with the standard non-filleted design in Fig. 22(a). The experimental results were then compared with the analytical solution to verify the force equations for each case [Eqs. (3), (14), (15), (17), and (24)] with their associated correction factors [Eqs. (25), (26), (27), and (28)], as shown in Figs. 23–27. The results show relatively good matching with the analytical solutions for each of the cases. All cases show an error matching of less than 11.6%, with the error

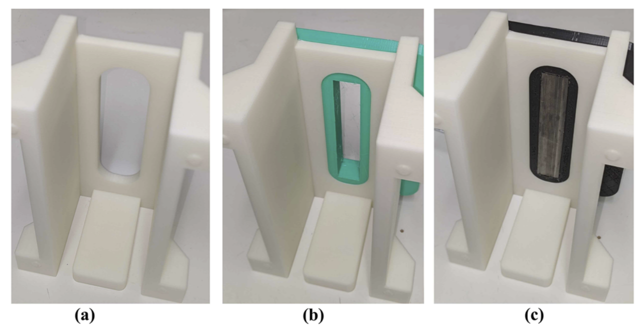

FIG. 21. The modular I-box design (a) allows for certain I-beam cases to be inserted into the back (b), making experimenting easier for each case. The I-beam can, then, be inserted (c) and held in place with pins.

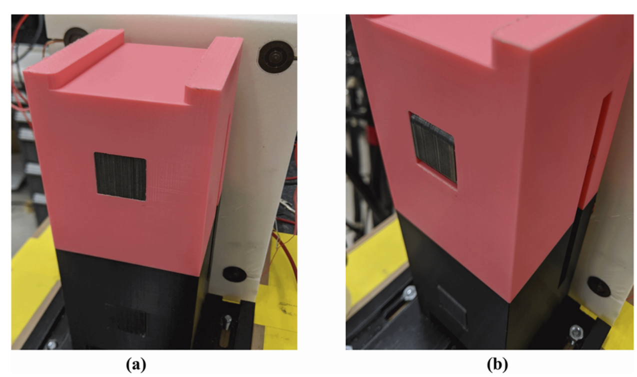

FIG. 22. The C-core setup (a) without fillets and (b) with fillets.

shown to relatively decrease at higher forces. Some sources of the potential error include systematic errors in the current measurements or force readings, 3D printing inaccuracies, hysteretic nonlinearities, flux fringing, and random errors from temperature variations.

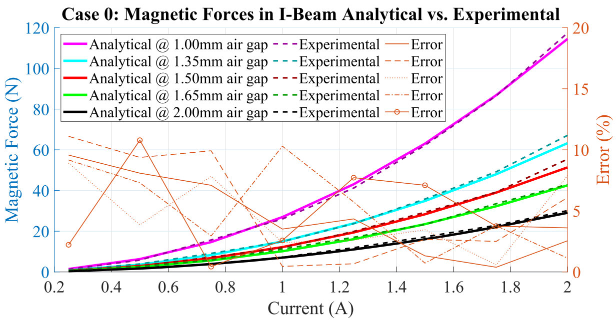

FIG. 23. The experimental results are shown for case 0 and compared to the analytical solution. The error is shown to be relatively low between the analytical and experimental model with a maximum error of 11.1%.

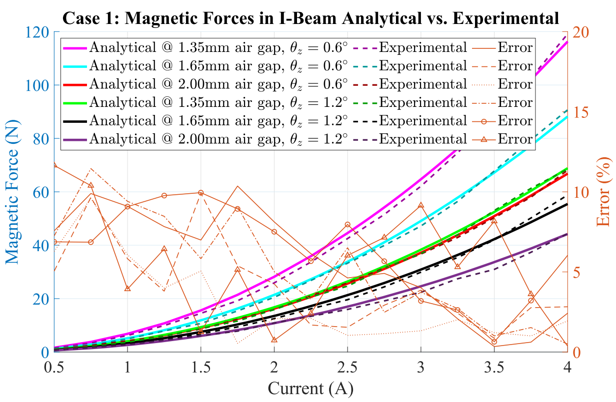

FIG. 24. The experimental results are shown for case 1 and compared to the analytical solution. The error is shown to be relatively low between the analytical and experimental model with a maximum error of 11.6%.

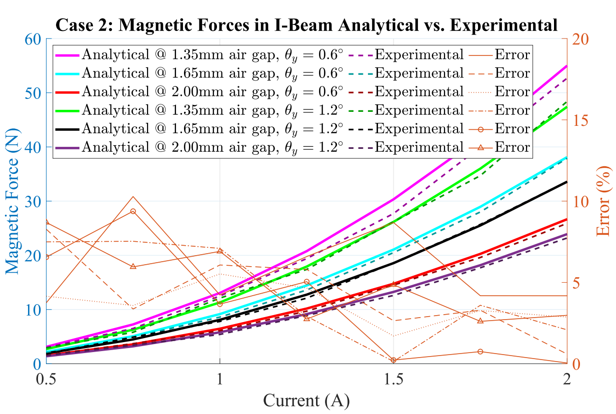

FIG. 25. The experimental results are shown for case 2 and compared to the analytical solution. The error is shown to be relatively low between the analytical and experimental model with a maximum error of 10.3%.

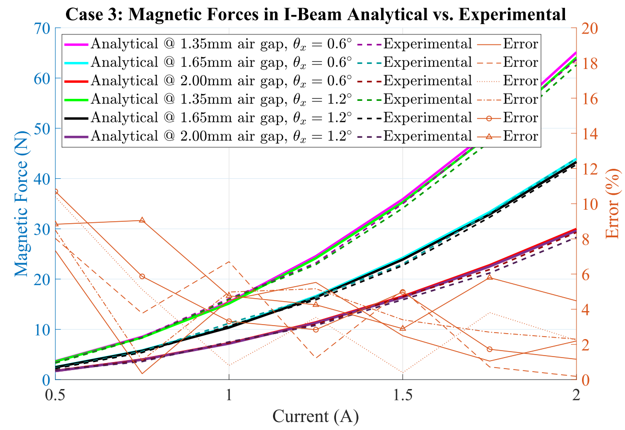

FIG. 26. The experimental results are shown for case 3 and compared to the analytical solution. The error is shown to be relatively low between the analytical and experimental model with a maximum error of 10.7%.

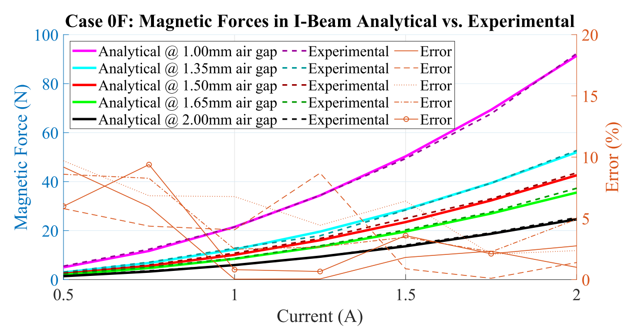

FIG. 27. The experimental results are shown for case 0 F and compared to the analytical solution. The error is shown to be relatively low between the analytical and experimental model with a maximum error of 9.7%.

V. DISCUSSION

The experimental model was found to match the analytical/FEA derived analytical equations for each case with an error below ~ 11%. One key takeaway from the experiment was that the C-box enclosure could only take so much force before elastic deformation occurred; therefore, all cases besides case 1 had to be limited to 2 A of current since the output forces were too high. A stronger material or a different design could extend the force range. Another takeaway is that case 3 is very similar to that of case 0 since angular offsets in the x axis has relatively small effects to the reluctance in the system, as shown in Ref. 26. For all cases, an increase in the angular offset decreases the overall output force of the reluctance actuator, as shown previously in Fig. 17. Case 1 is shown to have the greatest effect on the force since this case has the greatest effect on the governing reluctance. Overall, using derived analytical equations combined with known FEA simulation results allowed the force to be estimated within reasonable precision. The force sensors and displacement gauge used could also be improved for future experiments. In addition, another consideration would be to use experimental data instead of FEA results for the 3D mapping correction factors to better match the real world system. However, experimental data may have systematic errors depending on the setup and precaution should be made.

VI. CONCLUSION

The next evolution of the photolithography machine requires a more efficient actuation system, which can meet the high acceleration and precision demands in order to manufacture advanced semiconductor devices more efficiently and accurately. Currently, Lorentz actuators are utilized since they have a highly linear force–current dependency, which leads to easier modeling and control. However, Lorentz actuators require efficient cooling systems because of thermal/power limitations. Reluctance actuators, on the other hand, can provide a relatively high force output and are highly efficient although they are harder to model due to their nonlinear nature. If there are any disturbances such as asymmetries, the force output can be significantly changed. This paper highlighted a successful method to model the forces when the mover parts of the C-core reluctance actuator experience various angular disturbances or if non-typical modeling needs to be performed, such as a filleted design. The method uses data obtained from FEA models from the COMSOL software to correct the rough derivations for each disturbance case using polynomial curve fitting, which can predict the complex physics. The models were then found to match relatively well (within ~11%) with the experimental findings. The results will enhance the development of control systems for precision motion systems to overcome these asymmetrical air gap disturbances. Future studies on machine learning of the FEA data will be conducted to better predict various cases of disturbances in the system followed by a feedforward control to improve the motion control of reluctance actuated systems. In addition, multi-axis disturbances will be considered simultaneously such as having asymmetrical offsets in all angular degrees of freedom and a general analytical solution will be found.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Michael Pumphrey: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing-original-draft (equal); Writing-review-editing (equal). **Natheer Alatawneh:** Conceptualization (supporting); Investigation (supporting); Methodology (supporting); Project administration (equal); Supervision (equal); Validation (supporting); Visualization (supporting); Writing-original-draft (supporting); Writing-review-editing (supporting). **Mohammad Al Janaideh:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing-original-draft (equal); Writing-review-editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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